

Activity- 1:

Write a Python program to convert a list of numeric value into a one-dimensional NumPy array.

Activity- 2:

Create a 3x3 matrix with values ranging from 2 to 11.

Activity- 3:

Create a null vector of size 15 and update fifth value to 12 and ninth value to 17.

Activity- 4:

Take 2D and 3D array and perform transpose function.

Activity- 5:

Write a Python program to find the real and imaginary parts of an array of complex numbers.

Activity- 6:

Multiply a 5x3 matrix by a 3x2 matrix create a real matrix product using dot ().

Activity- 7:

Create array by 2x3 and perform following function:

- Number of dimensions
- Total number of element/size in an array
- What type of array stores in memory(dtype)

Activity- 8:

Write a Python program to create a 2-D array with reshape (5, 5) and filled 1 on the border and 0 inside using slicing method.

Activity- 9:

Write a Python program to create a 3-D array with ones on the diagonal and zeros elsewhere.

Activity- 10:

Write a Python program to find common values between two arrays using intersect1d (). Take array1= [1 30 40 60 80, 90] and array2= [50,10,30,40] as input.

Activity- 11:

Write a Python program to draw individual line charts of the financial data of a company given in Table A-11 between October 3, 2022 to October 7, 2022 using subplot for each attribute.

Table A-11: Financial data of a company

Date	Open	High	Low	Close
10-03-22	774.25	776.065002	769.5	772.559998
10-04-22	776.030029	778.710022	772.890015	776.429993
10-05-22	779.309998	782.070007	775.650024	776.469971
10-06-22	779	780.47998	775.539978	776.859985
10-07-22	779.659973	779.659973	770.75	775.080017

Activity- 12:

Write a Python program to draw line charts on the same graph of the financial data of a company given in Table A-11.

Activity- 13:

Write a Python program to draw scatter plot graph of the financial data of a company given in Table A-11.

Activity- 14:

Write a Python program to draw bar graphs for each attribute of the financial data of a company given in Table A-11.

Activity- 15:

Write a Python program to draw pie chart for each attribute of the financial data of a company given in Table A-11.